



# Business News Dataset from Google News API

## Introduction

This project performs Exploratory Data Analysis (EDA) on a news dataset collected from multiple publishers. The dataset contains information about news titles, snippets, publishers, and publishing timestamps. The purpose of this project is to analyze news publishing patterns, understand content distribution, and identify useful insights from large amounts of textual data.

During this project, several real-world data problems were identified and solved through data preprocessing and analysis techniques.

## Problems Identified in the Dataset

- Missing or incomplete values in some columns
- Different title and snippet lengths causing inconsistent text data
- Large amounts of unstructured news content
- Difficulty in understanding publishing trends directly from raw data
- Need for proper organization and visualization of news information

## Problems Solved During the Project

- Cleaned and processed the dataset using Python and Pandas
- Checked and handled missing values
- Analyzed title and snippet lengths using statistical methods
- Identified top publishers and publishing trends
- Created charts and graphs for better understanding of the dataset
- Explored relationships between textual features using scatter plots and heatmaps

## What I Learned From This Project

- Data cleaning and preprocessing techniques
- Exploratory Data Analysis (EDA) methods
- Data visualization using Matplotlib and Seaborn
- Text-based data analysis
- Trend analysis and pattern identification
- Handling real-world datasets in Google Colab using Python

## Recommended Data Policies

- Maintain structured and organized datasets
- Ensure consistent formatting of text data
- Remove duplicate and incomplete records
- Store accurate timestamps for tracking publishing activity
- Apply proper data validation techniques before analysis
- Use clear categorization and labeling for better reporting

Overall, this project improved practical knowledge of data analysis, visualization, and real-world news dataset handling using Python.

```
In [25]: import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
```

## Dataset Overview

The dataset contains news articles collected from different publishers. It includes information such as news titles, snippets, publishers, and timestamps.

This dataset helps analyze publishing activity, content distribution, and text-based patterns using Exploratory Data Analysis (EDA).

```
In [4]: df = pd.read_csv("/content/news_data.csv")
```

## What are the first five rows of the dataset?

```
In [5]: df.head()
```

| Out[5]:  | <b>publisher</b>   | <b>title</b>                                      | <b>snippet</b>                                    | <b>newsUrl</b>                                                                                                    | <b>timestamp</b>    |
|----------|--------------------|---------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>0</b> | WSJ                | Stock Market Today: Oil Slips, Dow Futures Ris... | Federal Reserve nominee Kevin Warsh travels to... | <a href="https://www.wsj.com/livecoverage/kevin-warsh-f...">https://www.wsj.com/livecoverage/kevin-warsh-f...</a> | 2026-04-21 13:45:00 |
| <b>1</b> | Bloomberg          | Apple Bets New CEO John Ternus Will Bring Back... | When Apple Inc. announced Monday that longtime... | <a href="https://www.bloomberg.com/news/articles/2026-0...">https://www.bloomberg.com/news/articles/2026-0...</a> | 2026-04-21 13:39:11 |
| <b>2</b> | CNBC               | Dow rises 200 points as Trump predicts a deal ... | U.S. stocks rose on Tuesday as President Donal... | <a href="https://www.cnbc.com/2026/04/20/stock-market-t...">https://www.cnbc.com/2026/04/20/stock-market-t...</a> | 2026-04-21 14:36:00 |
| <b>3</b> | CNN                | Retail sales post biggest jump in more than 3 ... | Americans saw large chunks of their wallets ge... | <a href="https://www.cnn.com/2026/04/21/economy/us-reta...">https://www.cnn.com/2026/04/21/economy/us-reta...</a> | 2026-04-21 12:35:22 |
| <b>4</b> | The New York Times | Oil Prices Recede and Stocks Gain as Investors... | Oil prices pulled back and stocks inched highe... | <a href="https://www.nytimes.com/2026/04/21/business/oi...">https://www.nytimes.com/2026/04/21/business/oi...</a> | 2026-04-21 07:39:39 |

What are the last five rows of the dataset?

```
In [6]: df.tail()
```

|           | <b>publisher</b>          | <b>title</b>                                       | <b>snippet</b>                                     | <b>newsUrl</b>                                                                                                    | <b>timestamp</b> |
|-----------|---------------------------|----------------------------------------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|------------------|
| <b>18</b> | TechCrunch                | CEO and CFO suddenly depart AI nuclear power u...  | The startup, co-founded by former U.S. Energy ...  | <a href="https://techcrunch.com/2026/04/20/fermi-ceo-an...">https://techcrunch.com/2026/04/20/fermi-ceo-an...</a> | 2026-04-20 15:30 |
| <b>19</b> | The Washington Post       | Saw a show through Ticketmaster? You may get a...  | The District reached a settlement with Live Na...  | <a href="https://www.washingtonpost.com/dc-md-va/2026/0...">https://www.washingtonpost.com/dc-md-va/2026/0...</a> | 2026-04-20 00:21 |
| <b>20</b> | statnews.com              | Eli Lilly's \$3.25 billion acquisition of Kelon... | Eli Lilly is spending \$3.25 billion to acquire... | <a href="https://www.statnews.com/2026/04/20/kelonia-th...">https://www.statnews.com/2026/04/20/kelonia-th...</a> | 2026-04-20 21:30 |
| <b>21</b> | Investor's Business Daily | Dow Jones Futures Fall; Trump Says Talks, Iran...  | Dow Jones futures fell as Iran said it has "no...  | <a href="https://www.investors.com/market-trend/stock-m...">https://www.investors.com/market-trend/stock-m...</a> | 2026-04-20 12:00 |
| <b>22</b> | MarketWatch               | The stock market's record rally now faces thes...  | A touch-and-go situation in the Strait of Horm...  | <a href="https://www.marketwatch.com/story/stock-market...">https://www.marketwatch.com/story/stock-market...</a> | 2026-04-20 18:40 |

What are the column names in the dataset?

```
In [7]: df.columns
```

```
Out[7]: Index(['publisher', 'title', 'snippet', 'newsUrl', 'timestamp'], dtype='object')
```

What is the shape of the dataset?

```
In [8]: df.shape
```

```
Out[8]: (23, 5)
```

What is the basic information of the dataset?

```
In [9]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 23 entries, 0 to 22
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0   publisher    23 non-null     object
1   title        23 non-null     object
2   snippet      23 non-null     object
3   newsUrl      23 non-null     object
4   timestamp    23 non-null     object
dtypes: object(5)
memory usage: 1.0+ KB

```

## What is the statistical summary of the dataset?

```
In [10]: df.describe()
```

```
Out[10]:
```

|               | publisher          | title                                             | snippet                                           | newsUrl                                                                                                           | timestamp           |
|---------------|--------------------|---------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>count</b>  | 23                 | 23                                                | 23                                                | 23                                                                                                                | 23                  |
| <b>unique</b> | 16                 | 23                                                | 23                                                | 23                                                                                                                | 23                  |
| <b>top</b>    | The New York Times | Stock Market Today: Oil Slips, Dow Futures Ris... | Federal Reserve nominee Kevin Warsh travels to... | <a href="https://www.wsj.com/livecoverage/kevin-warsh-f...">https://www.wsj.com/livecoverage/kevin-warsh-f...</a> | 2026-04-21 13:45:00 |
| <b>freq</b>   | 3                  | 1                                                 | 1                                                 | 1                                                                                                                 | 1                   |

## Are there any missing values in the dataset?

```
In [11]: df.isnull().sum()
```

```
Out[11]:
```

|                  |          |
|------------------|----------|
|                  | <b>0</b> |
| <b>publisher</b> | 0        |
| <b>title</b>     | 0        |
| <b>snippet</b>   | 0        |
| <b>newsUrl</b>   | 0        |
| <b>timestamp</b> | 0        |

**dtype:** int64

# Publisher Analysis

Which publishers published the highest number of news articles?

```
In [13]: df["publisher"].value_counts()
```

```
Out[13]:
```

|                                  | count |
|----------------------------------|-------|
| <b>publisher</b>                 |       |
| <b>The New York Times</b>        | 3     |
| <b>TechCrunch</b>                | 3     |
| <b>CNN</b>                       | 2     |
| <b>WSJ</b>                       | 2     |
| <b>Bloomberg</b>                 | 2     |
| <b>CNBC</b>                      | 1     |
| <b>Ars Technica</b>              | 1     |
| <b>Yahoo Finance</b>             | 1     |
| <b>Reuters</b>                   | 1     |
| <b>AP News</b>                   | 1     |
| <b>Electrek</b>                  | 1     |
| <b>The Guardian</b>              | 1     |
| <b>The Washington Post</b>       | 1     |
| <b>statnews.com</b>              | 1     |
| <b>Investor's Business Daily</b> | 1     |
| <b>MarketWatch</b>               | 1     |

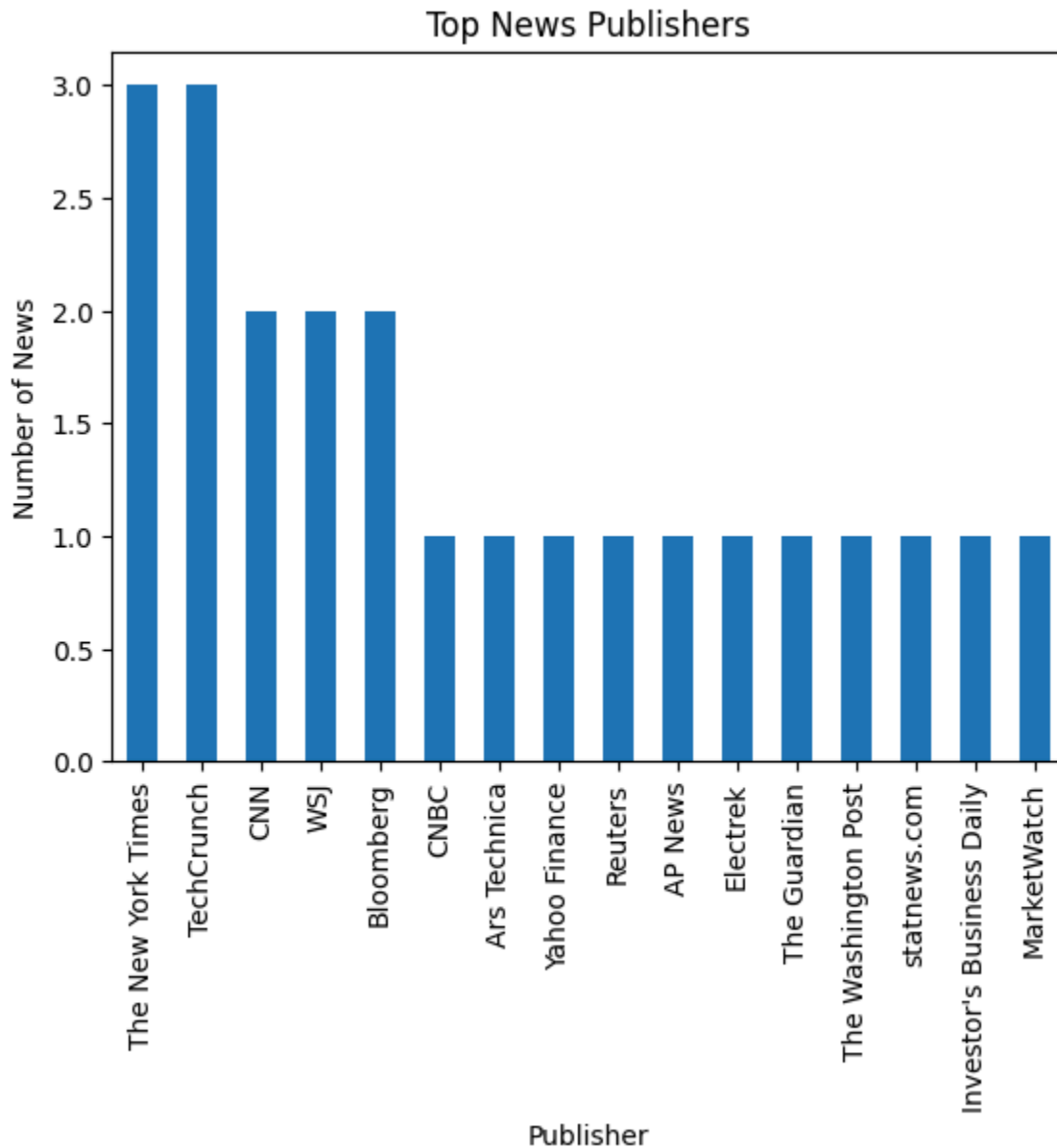
**dtype:** int64

## Top Publishers Bar Chart

```
In [15]: df["publisher"].value_counts().plot(kind="bar")

plt.xlabel("Publisher")
plt.ylabel("Number of News")
plt.title("Top News Publishers")
```

```
plt.show()
```



## Title Analysis

How many unique news titles are in the dataset?

```
In [16]: df["title"].nunique()
```

```
Out[16]: 23
```

# What is the distribution of news title lengths?

```
In [17]: df["title_length"] = df["title"].str.len()

df["title_length"].describe()
```

```
Out[17]:
```

|       | title_length |
|-------|--------------|
| count | 23.000000    |
| mean  | 74.260870    |
| std   | 16.979178    |
| min   | 30.000000    |
| 25%   | 63.500000    |
| 50%   | 72.000000    |
| 75%   | 88.500000    |
| max   | 111.000000   |

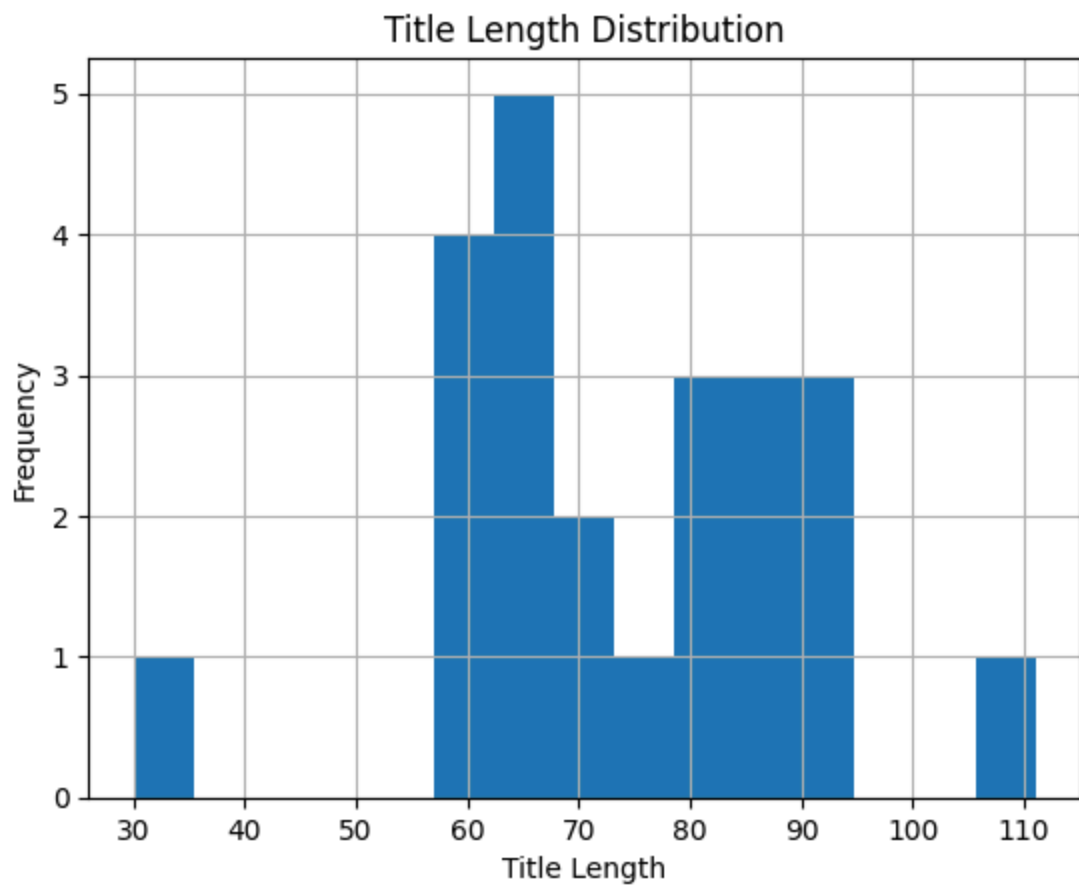
**dtype:** float64

## Title Length Distribution Histogram

```
In [18]: df["title_length"].hist(bins=15)

plt.xlabel("Title Length")
plt.ylabel("Frequency")
plt.title("Title Length Distribution")

plt.show()
```



## Snippet Analysis

What is the distribution of news snippet lengths?

```
In [19]: df["snippet_length"] = df["snippet"].str.len()  
df["snippet_length"].describe()
```

Out[19]:

| snippet_length |            |
|----------------|------------|
| count          | 23.000000  |
| mean           | 133.304348 |
| std            | 29.548781  |
| min            | 63.000000  |
| 25%            | 118.500000 |
| 50%            | 145.000000 |
| 75%            | 155.500000 |
| max            | 159.000000 |

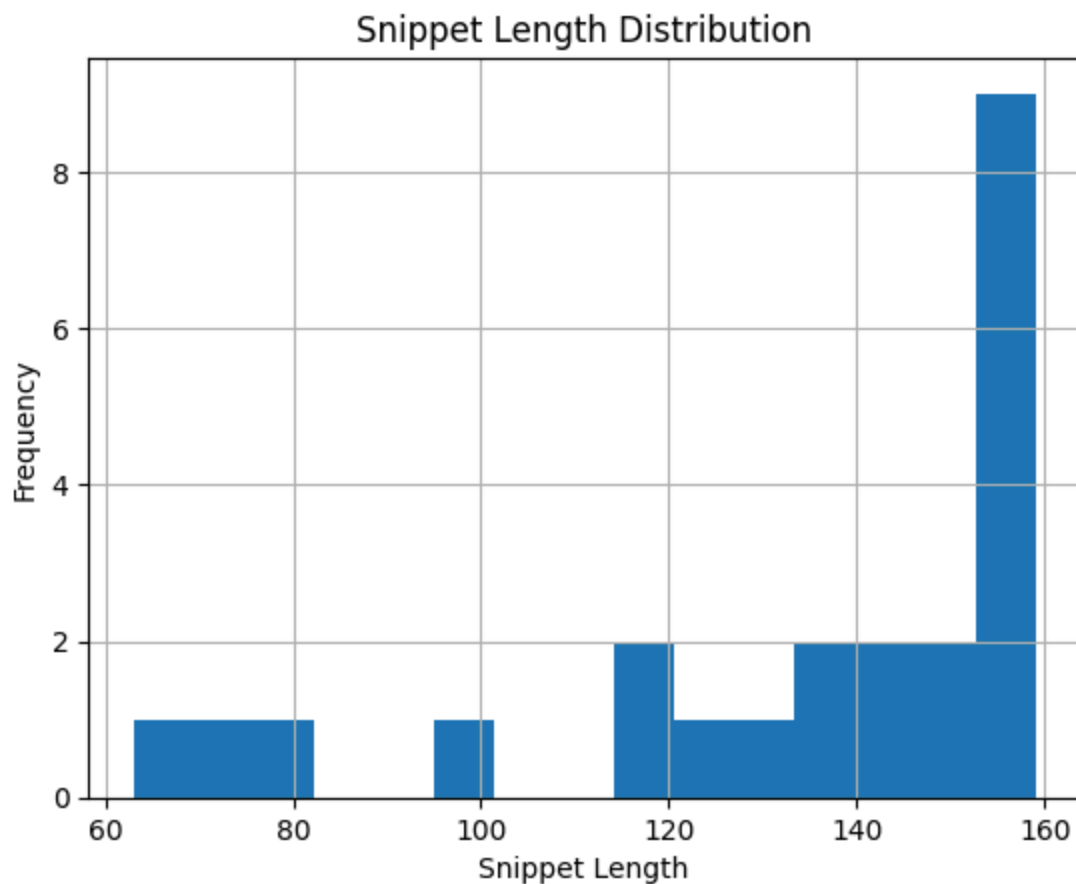
**dtype:** float64

## Snippet Length Distribution Histogram

```
In [20]: df["snippet_length"].hist(bins=15)

plt.xlabel("Snippet Length")
plt.ylabel("Frequency")
plt.title("Snippet Length Distribution")

plt.show()
```



## Timestamp Analysis

How many news articles were published each day?

```
In [21]: df["date"] = pd.to_datetime(df["timestamp"]).dt.date  
df["date"].value_counts()
```

```
Out[21]:
```

|            | count |
|------------|-------|
| date       |       |
| 2026-04-20 | 12    |
| 2026-04-21 | 11    |

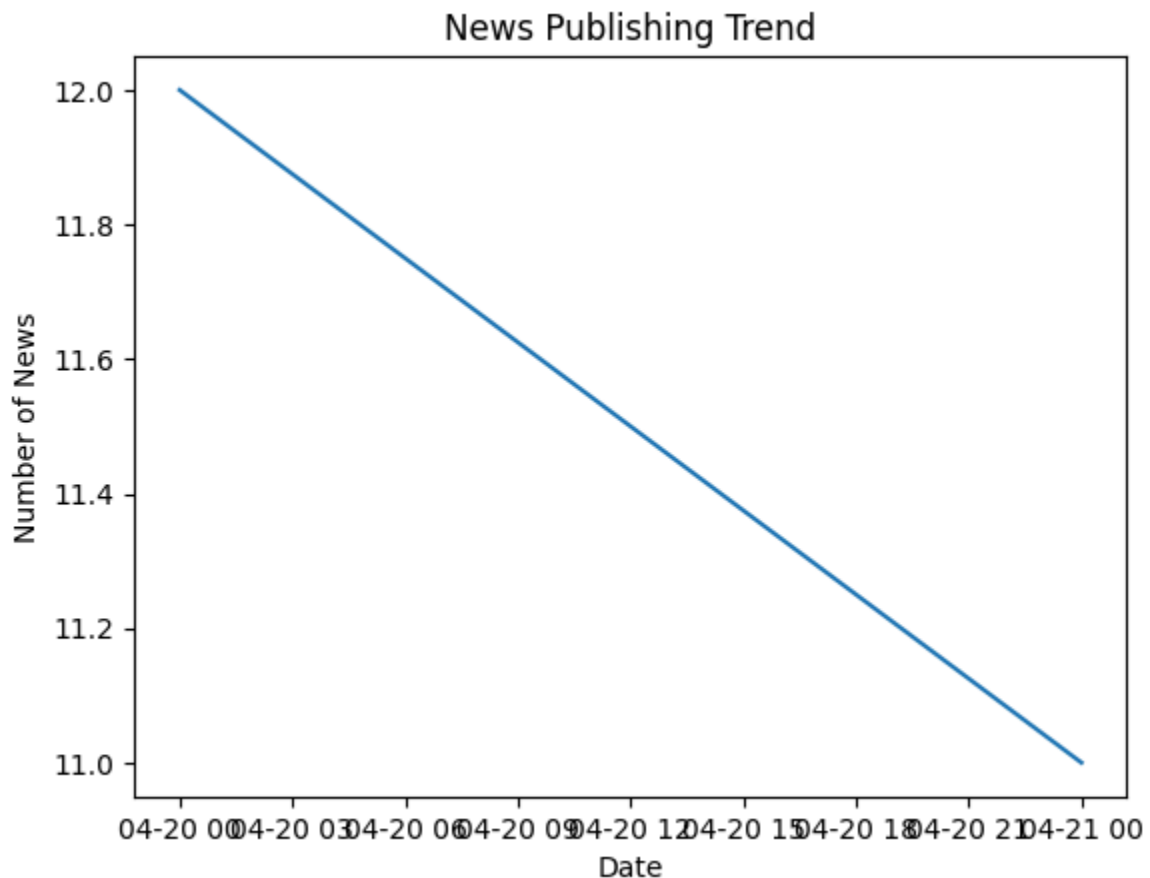
**dtype:** int64

# News Publishing Trend Line Chart

```
In [22]: df["date"].value_counts().sort_index().plot(kind="line")

plt.xlabel("Date")
plt.ylabel("Number of News")
plt.title("News Publishing Trend")

plt.show()
```



## Relationship Analysis

Is there any relationship between title length and snippet length?

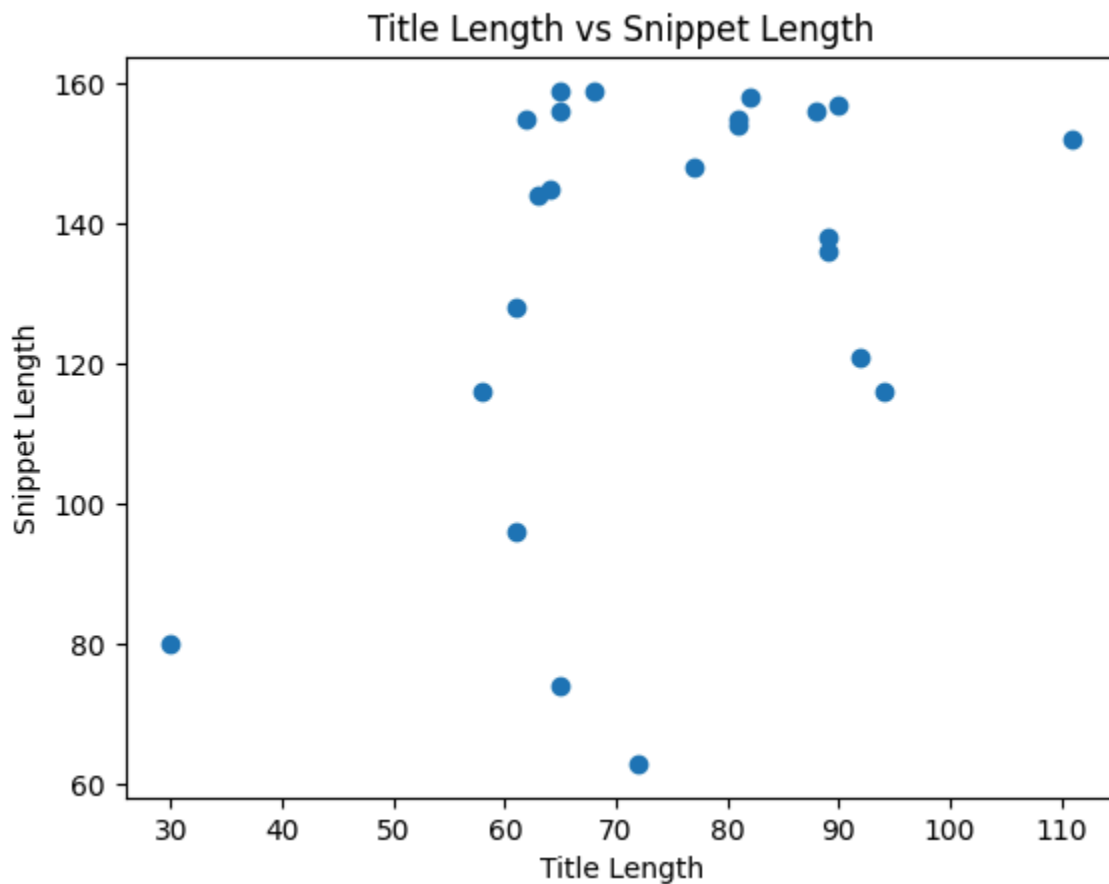
## Title Length vs Snippet Length Scatter Plot

```
In [23]: plt.scatter(df["title_length"], df["snippet_length"])

plt.xlabel("Title Length")
```

```
plt.ylabel("Snippet Length")
plt.title("Title Length vs Snippet Length")

plt.show()
```



What patterns can be observed from the correlation heatmap?

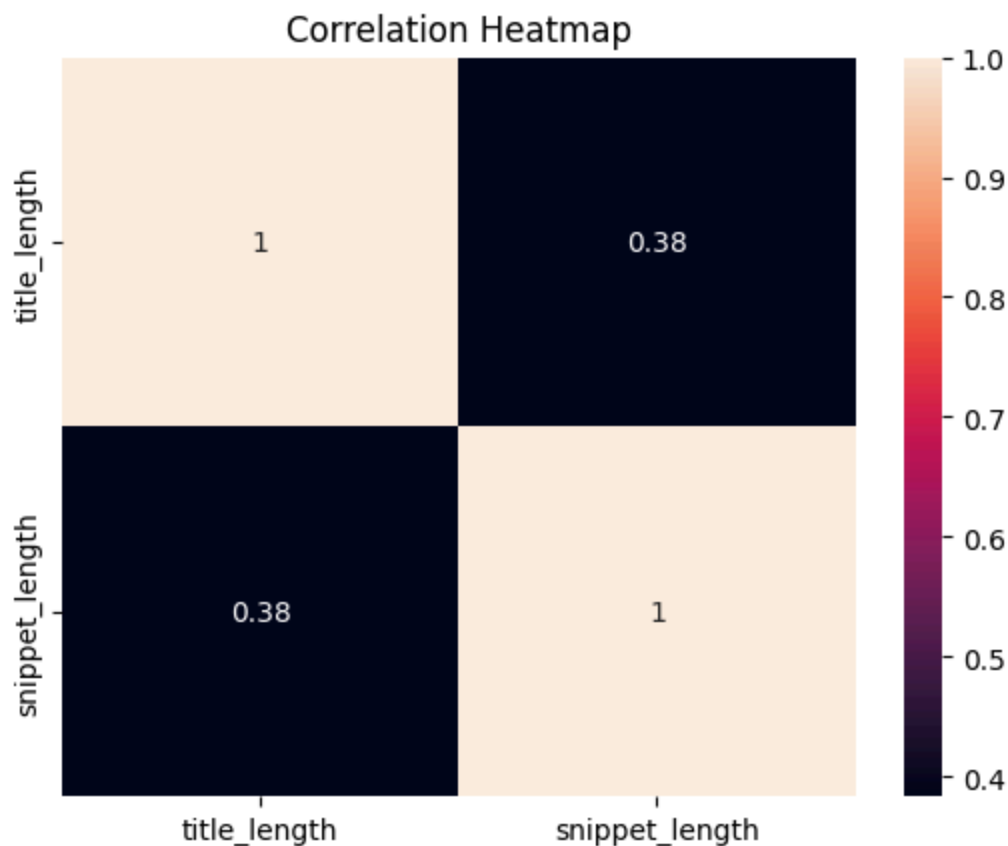
## Correlation Heatmap

```
In [24]: import seaborn as sns

sns.heatmap(df.corr(numeric_only=True), annot=True)

plt.title("Correlation Heatmap")

plt.show()
```



In [ ]:

In [ ]:

## Final Conclusion

This project successfully performed Exploratory Data Analysis (EDA) on a news dataset collected from multiple publishers. The analysis focused on understanding news publishing behavior, content distribution, textual patterns, and publisher activity using Python and data visualization techniques.

During the project, several challenges were identified in the dataset, including missing values, inconsistent text formatting, unstructured textual data, and difficulty in identifying publishing trends directly from raw information. These issues were addressed through data cleaning, preprocessing, statistical analysis, and visualization methods using libraries such as Pandas, Matplotlib, and Seaborn.

The analysis provided valuable insights into how news content is distributed across different publishers and how publishing activity changes over time. By analyzing title lengths, snippet lengths, publishing dates, and publisher contributions, the project revealed important patterns in news reporting and content organization.

Scatter plots and heatmaps were also used to identify relationships between textual features and numerical variables.

## Several visualizations were created during this project to better understand the dataset:

- Top Publishers Bar Chart was used to identify the publishers with the highest number of news articles.
- Title Length Distribution Histogram showed how news title lengths vary across the dataset.
- Snippet Length Distribution Histogram helped analyze the distribution of news snippet sizes.
- News Publishing Trend Line Chart was used to observe publishing activity over time.
- Title Length vs Snippet Length Scatter Plot helped identify relationships between textual features.
- Correlation Heatmap visualized relationships between numerical variables in the dataset.

This project also demonstrated the importance of proper data management and structured data policies. Maintaining clean and consistent datasets helps improve the accuracy of analysis and supports better decision-making. The project highlighted the need for proper text formatting, duplicate handling, timestamp accuracy, and organized data storage for effective analysis of large-scale news datasets.

## In addition, this project improved practical skills in:

- Data Cleaning and Preprocessing
- Exploratory Data Analysis (EDA)
- Statistical Analysis
- Data Visualization
- Text-Based Data Analysis
- Trend Identification
- Handling Real-World Datasets Using Python and Google Colab

Overall, the project successfully transformed raw news data into meaningful insights and visual representations. It provided a deeper understanding of publisher behavior, content structures, and news publishing trends while strengthening practical knowledge of data analysis and visualization techniques in

real-world scenarios.